

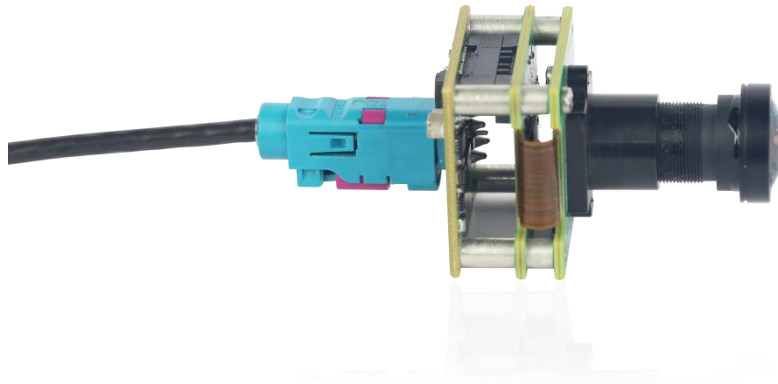


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NileCAM81



Module Datasheet

Revision 1.3

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1 Revision History

Rev	Date	Description	Author
1.0	22-Jul-2022	Initial draft	Camera Products Team
1.1	11-Jan-2023	Sensor format changed from 1/2" to 1/1.7"	Camera Products Team
1.2	30-Jan-2023	Updated frame rate	Camera Products Team
1.3	08-Jan-2024	Updated web and e-mail in the first page	Camera Products Team



2 Introduction

NileCAM81 is a family of NileCAM cameras from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81 offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81 fast (low latency), flexible and yet reliable.

NileCAM81 uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81 embeds the Maxim's serializer which serializes the high-speed image data and transmits over the coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81 recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81 is a 8 MP GMSL camera contains a 1/1.7" AR0821 CMOS image sensor from ON Semiconductor® along with on-board ISP. The coaxial cable is connected to Serializer Board through a rugged FAKRA connector. NileCAM81 can be interfaced with MAX9296A, MAX9296B and MAX96712 GMSL2 deserializers IC from Maxim's and also backward compatible with GMSL1 chips.

This document describes about the features of NileCAM81 mod and serializer board including the mechanical diagram.

3 Disclaimer

The specifications and features of NileCAM81 camera board are provided here as reference only and e-con Systems reserve the right to edit/modify this document without any prior intimation of whatsoever.

4 Description

NileCAM81 is a two-board solution. The first board is the standard NileCAM81 camera module that has 1/1.7" AR0821 image sensor with an electronic rolling shutter from ON Semiconductor® along with Image Signal Processor (ISP) and the second board is GMSL2 serializer board.

The following figure shows the various PCBs of NileCAM81.



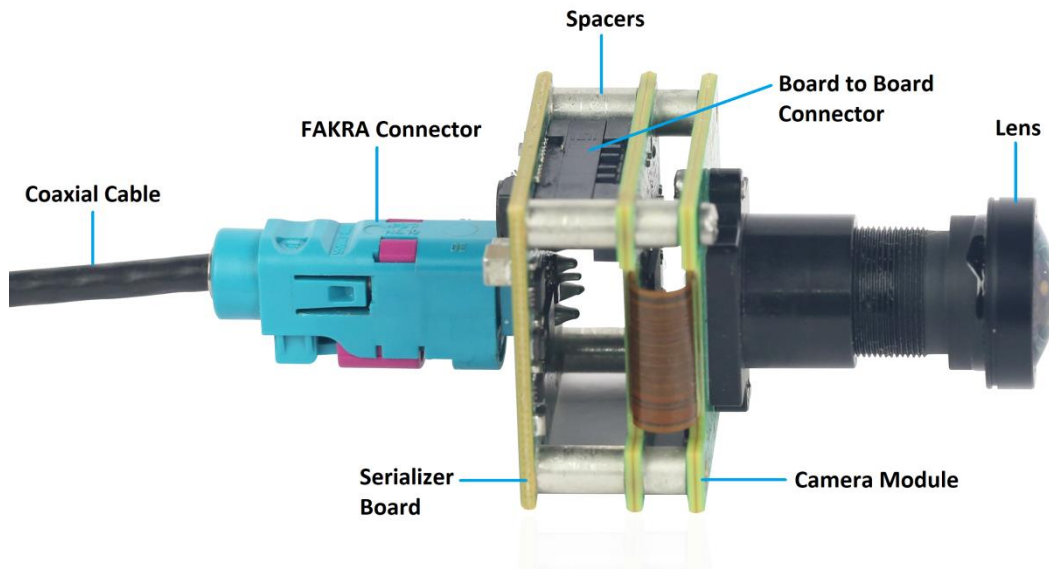


Figure 1: NileCAM81 Module

The camera board is based on AR0821, an 8 MP 1/1.7" CMOS image sensor from ON Semiconductor® along with ISP. The camera is available with M12 metal lens holder allowing customers to choose the lens as per their requirements. The serializer board is based on Maxim GMSL2 serializer chip and has all the circuitry required for bidirectional data communication, power and data separation. The FAKRA connector is used for the coaxial cable interface.

4.1 Features

The features of NileCAM81_CUOAGX are as follows:

- Multi-board solution
- AR0821 8 MP camera sensor with uncompressed UYVY format
- High Dynamic Range Imaging
- Compatible with Jetson™ AGX Xavier/Orin development kit
- Standard M12 lens holder for use with customized optics or lenses for various applications
- Light weight, versatile, and portable design
- Imaging applications
 - 8 MP CMOS image sensor with 1/1.7" optical form-factor
 - Still capture supported resolution: HD (1280 x 720), FHD (1920 x 1080), 4K (3840 x 2160)
 - Video streaming supported resolution: HD (1280 x 720), FHD (1920 x 1080), 4K (3840 x 2160)
- Maximum power consumed: 2.496W
- Operating temperature range: -30°C to 70°C
- RoHS compliant



4.2 CMOS Image Sensor Features

The following table lists the key specifications of NileCAM81.

Sensor Specification	
Type / Optical Size	1/1.7" Optical format CMOS image sensor
Resolution	8MP
Pixel Size	2.1 μm x 2.1 μm
Sensor Active Area	3848 H x 2168 V
Sensitivity RGB (Green)	15.4 ke-/lux*sec
SNR (max)	41.8 dB
Dynamic Range	> 140 dB

Table 1: CMOS Image Sensor Features

4.3 Maximum Frame Rate Supported

The following table lists the maximum frame rate supported by the camera module.

Resolution	1280 x 720 (HD)	1920 x 1080 (FHD)	3840 x 2160 (4K)
UYVY	60	60	16

Table 2: Maximum Frame Rate Supported

5 Power over Coax (PoC)

The NileCAM81 module contains a camera module board (with AR0821 image sensor and ISP) and a Serializer board. The power for these pair of boards is supplied from the Deserializer through PoC.

5.1 Power and Signal Flow in PoC

The NileCAM81_CUOAGX Serializer and Deserializer communicate using PoC technology. The PoC channel transmits serialized video channel from Serializer to Deserializer (forward channel) and a lower frequency control channel from Deserializer to Serializer (reverse/back channel) along with DC power.

The camera module and Serializer board are powered from Electronic Control Unit (ECU) containing Deserializer and processor. The filter on the ECU side is used to merge DC power with the video and communication data, while the filter on the camera side is to separate the DC power from the video and communication data and powers its internal circuitry.

The following figure shows the power and signal flow in the PoC.



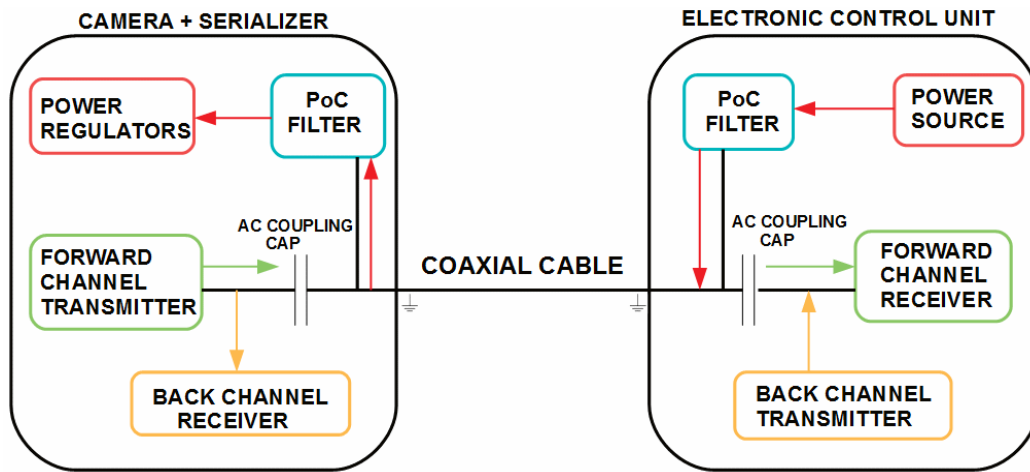


Figure 2: Typical PoC Power and Signal Flow

As a single coaxial cable transmits data with three different frequencies, a filter must be added on both sides of the communication link. Ideally the POC filter must have low impedance for DC signals and high impedance for back channel and forward channel signals. Hence, inductors are used for the PoC low pass filter construction.

Note: e-con Systems recommends 5V to 12V as standard voltage for NileCAM81 POC line.

5.2 PoC Filter Design

The PoC filter must have high impedance at back and forward channel frequencies to filter the DC power from the coaxial data. As 50Ω termination is used for data channels, a 20x increase in impedance ($1K\Omega$) is required to filter the high frequency content. Every inductor has parasitic capacitance that causes self-resonance and a corresponding drop in impedance at high frequencies. The inductors of different sizes are chosen to filter all the frequency bands of interest.

The following figure shows the series of inductors which are used to filter the high frequencies.

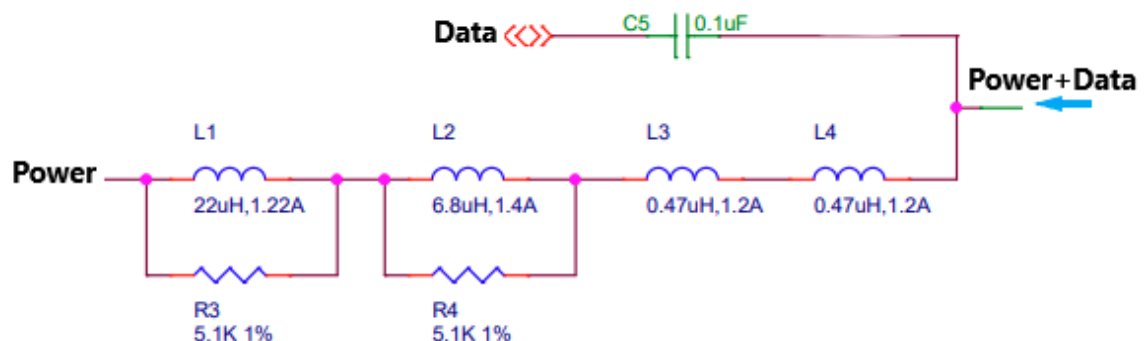


Figure 3: PoC Filter



As the PoC channel also contains required high frequency data, an AC coupling capacitor (C5) is placed near the NileCAM81 serializer to extract the data (filtering out DC).

5.3 PCB Layout Guidelines for PoC

The PCB layout guidelines for PoC are listed as follows:

- Route the coaxial trace to a FAKRA connector as a 50Ω single ended trace, with tight impedance control. If the trace is not 50Ω and laid out cleanly, performance issues can arise.
- Place the AC coupling capacitors close to the Serializer/Deserializer.
- Route high speed coaxial trace from the FAKRA connector with the most direct route and minimum trace length as possible. Also, the components on the path of this high-speed trace must be placed, to minimize the stub length seen by the transmission line.
- Ensure that the coaxial trace carries PoC current without temperature rise.
- Routing highest to lowest SRF inductor on the GMSL line is critical. Highest being positioned directly on the GMSL RF line followed by the next highest and so on.
- Minimize the distance between IC, PoC, and connector
- Cut out the ground on open coil inductors where the coil is exposed

Note: Reference Deserializer board schematics and PCB files can be shared based on request.

6 Connector Pinouts

The following sections describe the pins descriptions of NileCAM81 connector.

NileCAM81 has a high frequency FAKRA straight plug (CN2) with neutral coding, to mate with coaxial cable. The same FAKRA straight plug and pinouts are used in the Deserializer board. The connector on NileCAM81 is shown in the following figure.

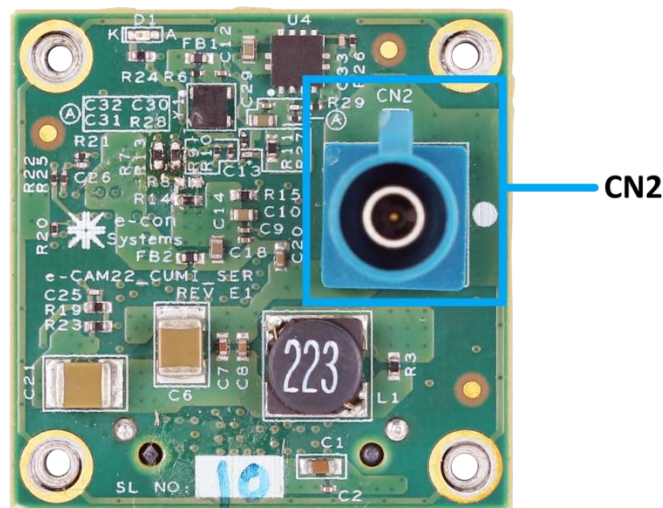


Figure 4: Connectors on Serializer Board



The following table lists the pin-outs details of FAKRA plug.

Pin No	Signal	Pin Type	Description
1	Coax	Input / Output	Coaxial signal that carries both power and bidirectional data
2	GND	Power	Ground
3	GND		Ground
4	GND		Ground
5	GND		Ground

Table 3: FAKRA Plug Pin-out Details

7 Connector Part Numbers

The following table lists the connectors used in the NileCAM81 camera board and its compatible mating connectors.

Connector	Manufacturer	Part Number
Coax Connector (CN2)	Rosenberger	59S10H-400T5-Z

Table 4: Connectors Part Numbers

As the NileCAM81 boards are available as two board solution, screws and spacers are used for holding the boards, at different points.

The following table lists the description of the part numbers.

Part No	Description	Manufacturer
1250388	Screws M2 Stainless steel 12 mm	Bossard

Table 5: Screws Part Number

Note: The above mechanical parts are verified for mechanical clearance with nearby components.

Warning: e-con Systems does not recommend using new mechanical screws and allied parts with the product. If new mechanical parts are to be used, care should be taken to avoid interference with nearby components.

8 Electrical Specification

This section lists the electrical specification and recommended operating conditions of NileCAM81.



8.1 Recommended Operating Condition

The following table lists the recommended operating condition of NileCAM81.

Parameter	Typical Operating Voltage	Current (mA)	Typical Power Consumption (W)
Streaming maximum power at 3840 X 2160 at 16fps	12V $\pm$ 250mV	208	2.496

Table 6: Recommended Operating Condition

Note:

- These values are measured in e-con Systems lab and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.
- The above current measured are for continuous device streaming.

8.2 Operating Temperature Range

The following table lists the operating temperature range of NileCAM81.

Temperature Range	Parameter Description
-30°C to 70°C	Electrically functional operating range

Table 7: Operating Temperature Range

This is the maximum temperature range up to which the camera sensor can be operated.

Note: The default lens supplied with this camera has an operating range of -30°C to 75°C. You can choose wider operating temperature lens as per your requirements. You are advised to make necessary arrangements on the products to dissipate the heat generated in the module to maintain the operating temperature below 70°C.

9 Mechanical Specifications

NileCAM81 is a two-board solution. The module and serializer boards have 30 mm x 30 mm form-factor and total height of this product is 52.21 mm. The height of the S-mount metal lens holder is 13 mm and the actual height of the module above the PCB depends on the selected lens.



The board dimensions are shown in the following listed sections:

- [Camera Module Board Dimensions](#)
- [Serializer Board Dimensions](#)
- [Lens Holder Dimensions](#)

9.1 Camera Module Board Dimensions

The front and rear views of NileCAM81 camera board mechanical details are shown in the following figures.

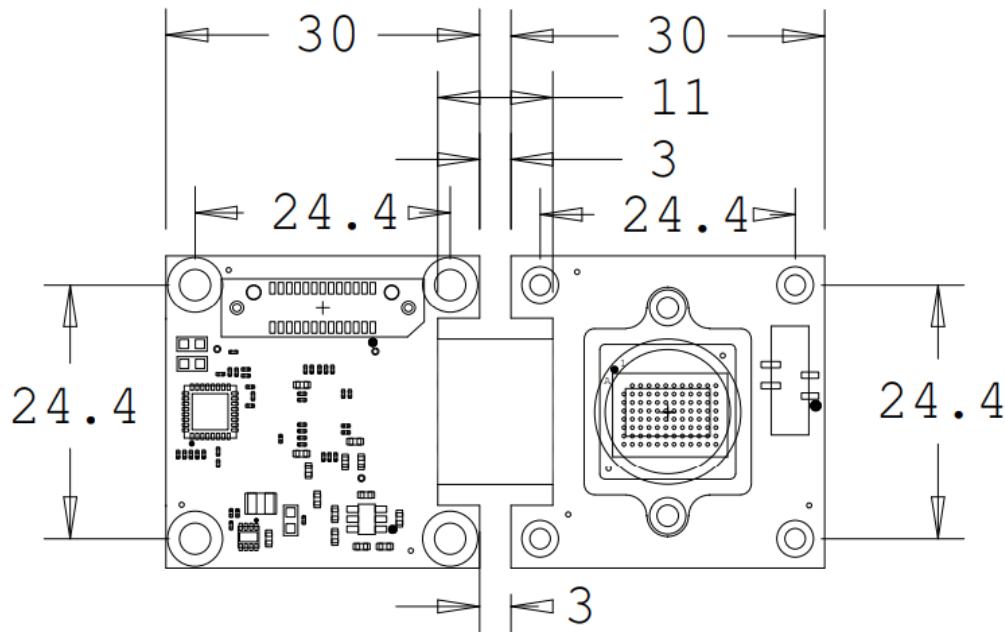


Figure 5: Top View of NileCAM81 Mod board Mechanical Drawing

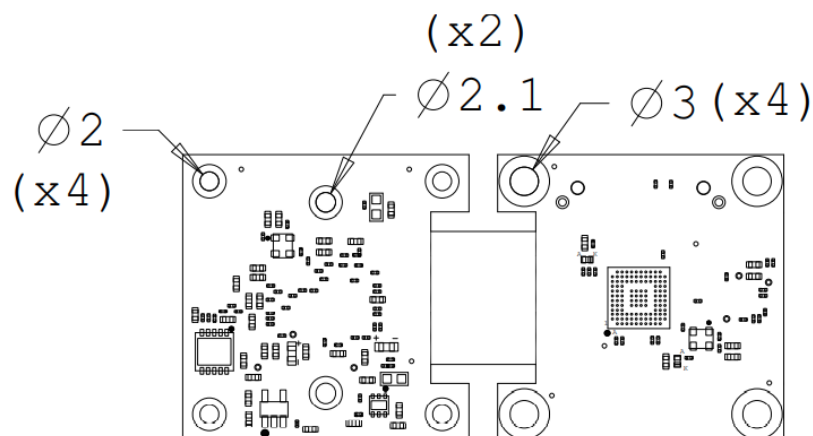


Figure 6: Bottom View of NileCAM81 Mod board Mechanical Drawing



Note:

1. All dimensions are in mm.
2. Board tolerance will be of +/- 0.2 mm

9.2 NileCAM81 Serializer Board Dimensions

The front and rear views of NileCAM81 Serializer board mechanical details are shown in the following figures.

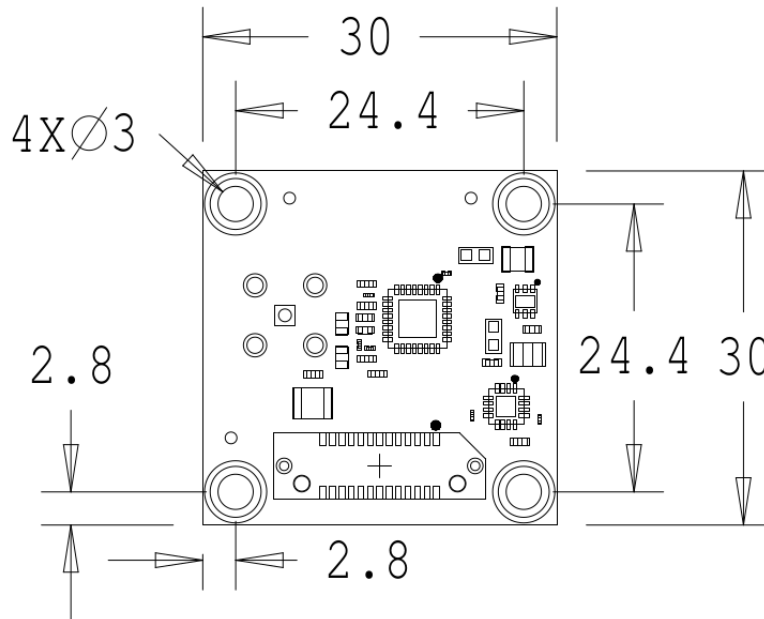


Figure 7: Front View of NileCAM81 Serializer Board

Note:

1. All dimensions are in mm.
2. Board tolerance will be of +/- 0.2 mm
3. NileCAM81_CUOAGX module board is two board solution connected with rigid flex cable. However, it will be given as folded condition



9.3 Lens Holder Dimensions

The following figure shows the dimension details of S-Mount metal lens holder.

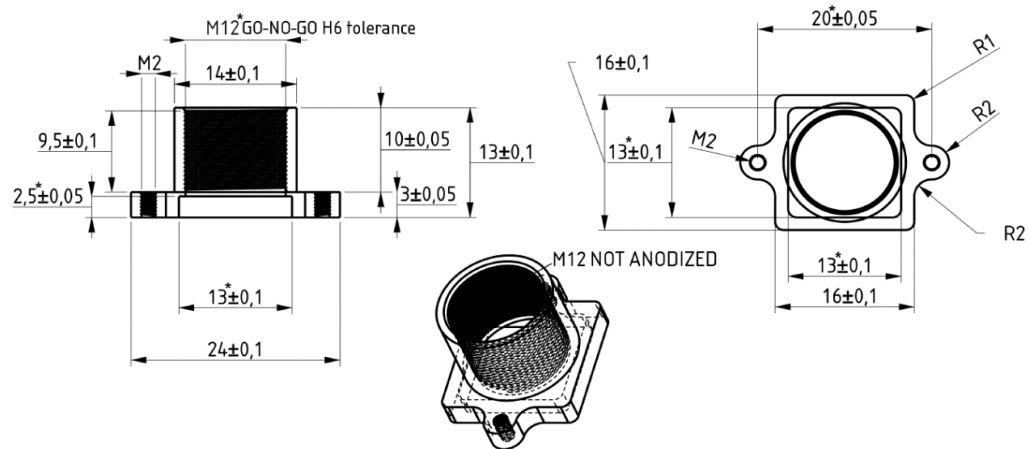


Figure 8: S-Mount Metal Lens Holder Outline Dimension

Note: All dimensions are in mm.

9.4 Mechanical Part Details

The mechanical accessories for e-CAM82_CUMI0821_MOD camera board are listed in following table.

Part	Quantity	Specification
Lens holder	1	Lens Holder M12 GO-NO-GO Thread Portion, Aluminium, 13mm Height, Centred with 20 mm Hole Spacing, Back Focal Length (BFL) 2.50mm
Lens holder screw	2	Screw M2 machine screw stainless steel head diameter 4mm max, thread diameter 2mm max, length 5mm

Table 8 : Mechanical Part Details



Support

Contact Us

If you need any support on NileCAM81 product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

